Math 306 Midterm 2 Review

Definitions to know: a T-invariant subspace, diagonalizable linear map, eigenvalue and eigenvector for T, isomorphic vector spaces, isomorphism, minimal polynomial, inner product, norm, orthogonal

Sample problems:

1. Prove that $\mathcal{L}(V)$ has a basis consisting of diagonalizable linear maps.

2. Prove that if $T^n = I$ for some positive integer n, then every eigenvalue λ of T satisfies $|\lambda| = 1$.

3. Proof or counterexample: If T^{-1} exists, then T is diagonalizable.

4. Prove that $T \in \mathcal{L}(V)$ is a scalar multiple of the identity map if and only if $ST = TS$ for every linear map $S \in \mathcal{L}(V)$.

5. Find a basis for which $T \in \mathcal{L}(V)$ defined by $T(a, b, c) = (a + b, a + c, a + c)$ is upper triangular. Can T be diagonalized?

6. Let $U = \text{span}\{e^x, e^{-x}\}$ be a subspace of the vector space of functions from $\mathbb{C}$ to $\mathbb{C}$. Define a linear map $T : U \to U$ by $T(f) = f - 3f'$. Find all eigenvalues of T.

7. For $T \in \mathcal{L}(\mathbb{C}^n)$, define $\exp T$ to be the linear map

$$\exp T = \frac{1}{0!}I + \frac{1}{1!}T^1 + \frac{1}{2!}T^2 + \cdots .$$

(Note: This definition makes sense only if the infinite series $(\exp T)v$ converges for all $v \in \mathbb{C}^n$; take this as a given.) Prove that T and $\exp T$ have the same eigenvectors.

8. Prove that if x, y are orthogonal nonzero vectors in a vector space V, then $\left\| \frac{x}{\|x\|} + \frac{y}{\|y\|} \right\|^2 = 2.$

9. Let u be a vector in an inner product space V. Proof or counterexample: $\{v \in V : u \text{ and } v \text{ are orthogonal}\}$ is a subspace of V.